

Mohammad Fardeen Shaik

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AI SOFTWARE ENGINEER

Education

George Mason University, College of Engineering

Aug 2025 – May 2027

M.S. Computer Science

Coursework: Intro Artificial Intelligence, Software Eng for WWW, Component based Software Development, Analysis of Algorithms, Computer Systems and System Programming, Mathematical Foundations of CS, Database Systems, Mobile Immersive Computing, Modern Computer Architecture

Vellore Institute of Technology

Sept 2021 – May 2025

B.Tech. Computer Science and Engineering

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, AI & ML

Technical Skills

Languages: Python, JavaScript, Java, SQL, R, HTML, CSS

Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebooks

Core Areas: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Statistics, Data Analysis, Feature Engineering

Professional Experience

Quadrant Technologies | Cloud AI Intern | Redmond, WA

Jun 2026 – Aug 2026

- Developed an explainable candidate ranking system using semantic search and LLMs with adjustable scoring weights across resume; implemented blind review mode showing exactly where each matched skill, ensuring full transparency trust in AI decisions.
- Built Screening Assistant using React, FastAPI, and Azure services to automate resume parsing; leveraged Azure AI Document Intelligence and Azure OpenAI for intelligent extraction enabling recruiters to reduce manual screening time significantly.

Ethnus Codemithra | Full-Stack Developer Intern

Aug 2023 – Nov 2023

- Enhanced UI/UX by refactoring reusable React components and optimizing state management, resulting in 15% increased user engagement; implemented automated tests with Jest/React Testing Library, API monitoring Postman and achieved 80% test coverage.

Projects

AI Contract Invoice Intelligence System | Python, TensorFlow, Apache Airflow, Azure

Nov 2025

- Built scalable NLP pipelines using Python and TensorFlow to automate invoice extraction and data enrichment from unstructured PDFs; successfully processed 150,000+ invoices annually with 98% accuracy, verified through quarterly QA audits.
- Engineered Apache Airflow DAG-based orchestration workflow covering end-to-end ingestion, text extraction, validation, and storage pipelines; implemented automated validation rules and error detection that reduced data quality issues by 60%.

Personalized Recommendation System | Python, FastAPI, React, Scikit-Learn (SVD, KMeans), Pandas, NumPy, SciPy

Oct 2025

- Engineered a multi-model recommendation platform leveraging SVD/Matrix Factorization, KMeans clustering, and RFM segmentation algorithms; implemented comprehensive evaluation metrics with popularity baselining to validate personalization uplift.
- Built a RESTful FastAPI backend with versioned endpoints for product recommendations, customer segmentation, and association rules; developed a decoupled React analytics dashboard displaying ML model outputs, customer insights and inventory intelligence.

Loan Default Prediction | PyTorch, XGBoost, SHAP/LIME, Streamlit

Feb 2026

- Developed an ensemble machine learning model combining PyTorch neural networks and XGBoost gradient boosting to predict loan defaults; performed extensive feature engineering, hyperparameter tuning, and cross-validation to achieve high predictive accuracy.
- Deployed an interactive Streamlit web application powered by SHAP and LIME explainability frameworks, enabling non-technical banking professionals to understand model predictions and make transparent, data-driven lending decisions with full audit trails.

Autism Detection | Python, TensorFlow, CNN/LSTM, Flask, MySQL

Apr 2026

- Designed and trained CNN/LSTM hybrid deep learning architecture for autism spectrum disorder screening; enhanced baseline model accuracy by 15% to achieve 91% using advanced training techniques learning-rate scheduling, and class-weight balancing.
- Built end-to-end deep learning pipeline with data preprocessing, model training, validation, and evaluation on 1,500+ labeled medical samples; integrated Flask backend with MySQL database for patient data management and integrated model inference.

Research

Hybrid Algorithm for Parkinson's Prediction | Python, Scikit-Learn (Random Forest, SVM), SQL.

Nov 2025

- Presented hybrid machine learning model at Springer 2024 combining Random Forest and SVM algorithms, achieving 88.7% accuracy on UCI Parkinson's dataset (195 instances) with 6-8% improvement over individual models by optimizing sensitivity and specificity for early-stage neurological disorder detection.

Leadership

Secretary of CS Graduate Student Association at George Mason University (Apr 2026 – Present):

- Drive administrative efficiency and member engagement by documenting organizational activities, ensuring compliance with university policies, and collaborating with leadership to execute talks, meetings, and networking events that strengthen the graduate CS community.